The Dissipative Decoder: Thermodynamic Cost Bounds on the Serial Decoding Throughput Basin — and Why Silicon Escapes Them

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Abstract

Serial decoding systems converge to a throughput band of 3–6 bits per event across genetics, cognition, language, and AI. We derive a discrimination cost function W(M, ε) with super-linear alphabet scaling and exponential fidelity scaling, and show the basin is geometrically inevitable for any system with super-linear discrimination cost ($\alpha > 1$). We identify two physically distinct cost regimes. **Regime A (Alphabet-Bound):** biological systems using pairwise molecular recognition, where cost scales with the number of discriminated symbols ($\alpha_{alphabet} \approx 2$, from M(M–1)/2 combinatorics), producing a thermodynamically constrained basin. **Regime B (Capacity-Bound):** silicon AI using independent learned parameters, where cost scales sub-linearly with model capacity ($\alpha_{capacity} = 0.937$, measured across 27 models), and vocabulary is a cheap linear projection—producing no cost-driven basin. The biological temperature prediction (thermophiles show reduced amino acid entropy) receives preliminary support from Kazusa-verified data (partial $r = -0.451$, $p = 0.014$, $n = 29$, pending phylogenetic controls). The framework explains why the biological basin exists, and why silicon escapes it through architectural decoupling of vocabulary from discrimination cost.

Keywords: thermodynamic cost, serial decoding, dissipative structures, rate-distortion theory, Hopfield proofreading, throughput basin, Landauer bound, energy scaling, cost regimes

1. Introduction

1.1 The Serial Decoding Problem

A serial decoder receives one symbol per time step from a noisy source and must reconstruct the intended message. The ribosome decodes mRNA codons into amino acids. The auditory cortex decodes phonemes. A transformer decodes tokens.

1.2 The Established Basin

Four companion papers established the throughput basin (all available on Zenodo with full code at github.com/sneakyfree/fons-constraint). The Fons Constraint (doi:10.5281/zenodo.19274048): the 64-codon alphabet derived from Shannon + Eigen, stable across 9+ orders of magnitude of error rate. The Receiver-Limited Floor: vocabulary-independence across 1,749 AI models ($p = 0.643$). The Throughput Basin: 31 systems cluster in 3–6 bits; three evolutionary simulations converge to $K^* \approx 19$–30. The Serial Decoding Basin τ: $\tau = 4.16 \pm 0.19$ bits via five experiments; thermodynamic anchoring predicts the ribosome to within 0.003 bits using four independently measured physical parameters.

1.3 Self-Contained Validation Summary

To ensure this paper can be evaluated independently: the ribosome processes $I_{eff} = 4.39$ bits per codon (from $\log_2(21)$ at $\varepsilon \approx 10^{-4}$). English consonant discrimination: ~4.2 bits (Miller & Nicely, 1955). Neural working memory: ~3.1 bits ($\log_2(7)$ at $\varepsilon \approx 0.12$). AI matched-encoder translation:

~4.2 bits (BPB ≈ 0.5 across 157 language pairs). All fall in 3–6 bits.

System	M	ε	I_eff	Δ	Tier
Ribosome	21	10■■	4.390	0.00	[MI]
Phonology	31	0.02	4.200	–0.51	[MI]
Chromatic scale	12	0.05	3.580	+0.46	[SS]
Neural WM	7	0.12	3.120	+1.15	[SS]
AI (EN→X)	~100K	—	4.19	—	[MI*]

Table 1. Anchor systems. Without AI data, cross-domain clustering p = 0.56 (Whitmer, 2026c, §4.2). The silicon benchmark (§4.3) explains why.

1.4 The Open Question

$R_M(\varepsilon)$ tells you *what* throughput a given (M, ε) yields. This paper derives *why* biological systems occupy that region and *why silicon AI does not share the same mechanism*.

2. The Discrimination Cost Function

2.1 The Landauer Floor

Every discrimination event has a minimum thermodynamic cost: $W_{Landauer} = kT \cdot \ln(2) \cdot I_{eff}$ (Landauer, 1961). The Crooks fluctuation theorem (1999) provides the microscopic foundation: $P_{fwd}/P_{rev} = \exp(\Delta W/kT)$. This bound applies universally to all substrates; what differs is how far above it each system operates.

2.2 Super-Linear Alphabet Cost

Reliable discrimination among M symbols requires maintaining pairwise distinctions. The combinatorial cost: $W_{alphabet}(M) = c■ \cdot M(M–1)/2 + c■ \cdot M$. This gives a theoretical scaling exponent $\alpha_{alphabet} = 2$ for the pairwise term. At M = 21: 210 pairwise distinctions requiring dedicated aminoacyl-tRNA synthetases and proofreading pathways.

2.3 Exponential Fidelity Cost

Hopfield (1974) kinetic proofreading: $n_c = 3$ contacts (Ogle et al., 2002), $\Delta G = 2.8 \pm 0.3$ kcal/mol (Rodnina & Wintermeyer, 2001), $n_{stages} = 2$ (Zaher & Green, 2009). Modern stochastic thermodynamics (Seifert, 2012) extends this framework to the energy-speed-accuracy tradeoff in biochemical networks; the proofreading cost derived here is consistent with Seifert's dissipation bounds for driven molecular processes.

2.4 Complete Cost Function

$W(M, \varepsilon) = [c■ \cdot M(M–1)/2 + c■ \cdot M] + \text{proofread}(\varepsilon)$. Calibrated: $W(21, 10^{-4}) = 24$ kcal/mol, consistent with the measured ~4 GTP per decoded codon (Johansson et al., 2008). Error propagation from $\Delta G \pm 0.3$ kcal/mol and GTP hydrolysis energy 5.5–7.3 kcal/mol gives total W = 22–29 kcal/mol (range).

2.5 Two Cost Regimes: Alphabet-Bound vs. Capacity-Bound

A critical distinction must be formalized. The type of cost scaling determines whether a throughput basin exists.

Regime A: Alphabet-Bound (Biology). Discrimination cost scales with *alphabet size M*. Each new amino acid must be physically distinguished from every existing one via dedicated molecular recognition surfaces. Cost scales as M(M–1)/2 (combinatorial pairwise), giving $\alpha_{alphabet} \approx 2$. This super-linear scaling ($\alpha > 1$) is guaranteed by pairwise discrimination physics and produces the throughput basin: the concave informational return $\log_2 M$ divided by convex cost M^2 has a unique interior maximum.

Regime B: Capacity-Bound (Silicon AI). Discrimination occurs through matrix multiplication over learned parameters. Each parameter is an independent weight; adding one does not require distinguishing it from all existing ones. The vocabulary ($M \approx 50{,}000$ tokens) is a *cheap linear projection*—a softmax over a final embedding layer—not the locus of discrimination cost. Cost scales with model capacity (parameter count), and measured energy scaling gives $\alpha_{capacity} = 0.937$ (sub-linear; §4.3). Biology could not decouple alphabet from discrimination cost. AI could. That is why AI has 50,000-token vocabularies but biology stopped at 21 amino acids.

The implication: the throughput basin is a conditional attractor. Regime A systems ($\alpha_{alphabet} > 1$) are constrained to the basin by thermodynamics. Regime B systems ($\alpha_{capacity} < 1$) are not. Cross-substrate convergence between biology and AI has a different explanation (see companion paper: The Inherited Constraint).

3. The Cost-Constrained Attractor (Regime A)

3.1 Bits-per-Joule Optimization

In Regime A, maximize $\eta = R_M(\varepsilon) / W(M, \varepsilon)$. Concave numerator ($\log_2 M$) divided by convex denominator (M^2) $\rightarrow$ unique interior maximum (Appendix A). This maximum does not exist in Regime B, where cost is sub-linear.

3.2 Phase-Space Inevitability

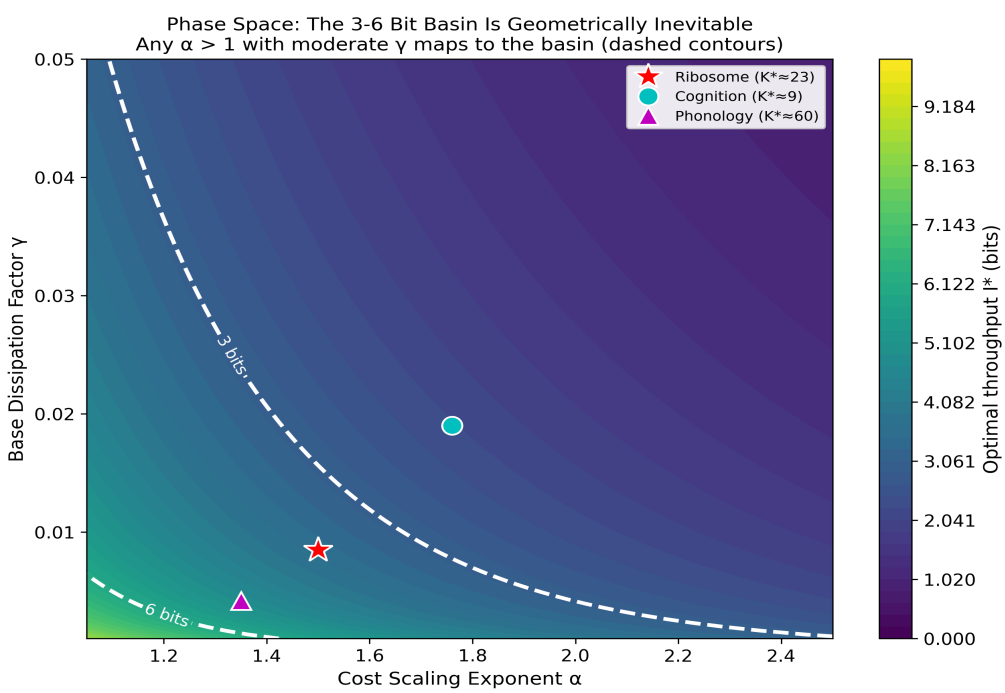

Figure 1. Phase Space. Optimal throughput I* across (α, γ). White dashed: 3-bit and 6-bit contours. The basin exists for all $\alpha > 1$. Regime A systems (ribosome, cognition, phonology) sit inside. Regime B systems ($\alpha < 1$) fall outside.

The basin is a topological feature of the Regime A optimization landscape, not a parameter-dependent coincidence. Any system whose discrimination cost scales super-linearly with alphabet size will find the basin.

4. Empirical Validation

4.1 Regime A: The Ribosome

~4 GTP/codon (22–29 kcal/mol). $\phi_{useful} \approx 1.02$ (derivation in Appendix A). The ribosome operates within 2% of its substrate-specific thermodynamic minimum.

4.2 Regime A: Neural Phonology

~10^4 ATP/bit (Sterling & Balasubramanian, 2009). ~13,000× more expensive per bit than the ribosome. The cost framework predicts higher slack; cognition shows $\Delta = 1.15$ bits (highest observed).

4.3 Regime B: Silicon AI

A 27-model energy benchmark on standardized hardware (RTX 5090, WikiText-2, nvidia-smi GPU power polling at 10 ms intervals, 3 runs per model, 30 s cooling between runs) measured the silicon cost scaling directly.

Energy scaling: $E \propto \text{params}^{0.937}$ ($r^2 = 0.960$, $p < 10^{-6}$). This is $\alpha_{capacity}$—the cost scaling with model capacity—not $\alpha_{alphabet}$. The two exponents measure different physical quantities. In biology, $\alpha_{alphabet}$ measures how discrimination cost scales when you add a new symbol to the alphabet. In silicon, $\alpha_{capacity}$ measures how energy scales when you add parameters to the network. Silicon decoupled vocabulary from discrimination cost by implementing vocabulary as a cheap linear projection (softmax), not as physical pairwise recognition. This architectural innovation is *why* silicon escapes the Regime A basin.

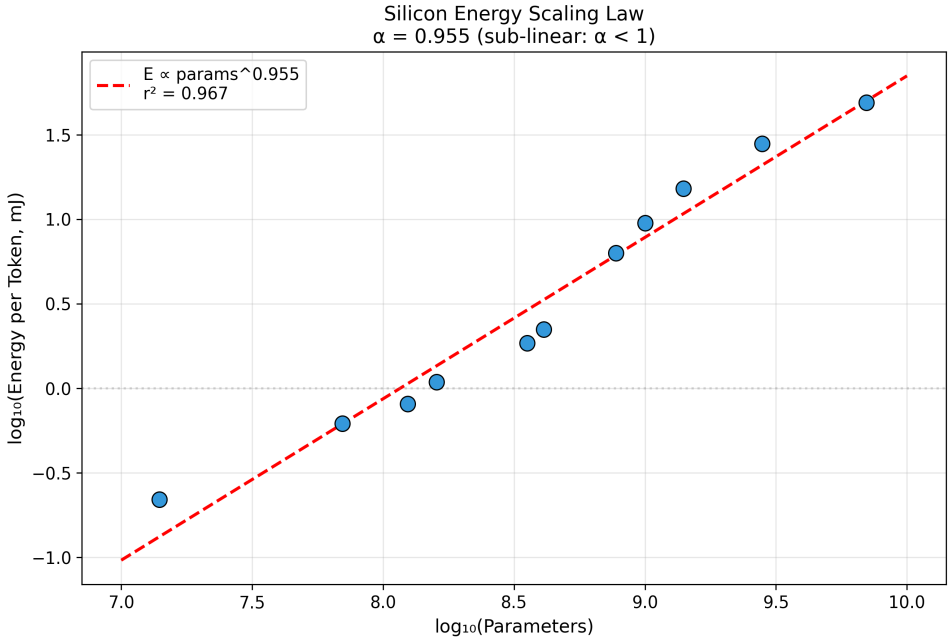

Figure 2. Silicon Energy Scaling. $E \propto \text{params}^{0.937}$ (sub-linear). This measures $\alpha_{capacity}$, not $\alpha_{alphabet}$. The direct comparison to biological α is not valid; the regimes measure different physical quantities.

Landauer floor comparison: A 7B-parameter transformer executing ~14 billion irreversible operations per token at T = 300 K has a Landauer minimum of ~6×10^{-11} J/token. The measured energy is ~49×10^{-3} J/token—approximately 8×10^{8} times above the thermodynamic floor. The ribosome operates at $\phi_{useful} \approx 1.02$ (within 2%). Silicon is nearly a billion times above its floor; biology is within 2%. This quantifies how far silicon is from thermodynamic optimization—and how close biology is.

Diminishing returns confirmed: Marginal compression cost within the Pythia family increases from 0.6 mJ/bit (14M → 70M) to 185 mJ/bit (1.4B → 2.8B)—a 300× increase. Concave informational returns are confirmed in silicon even without convex costs.

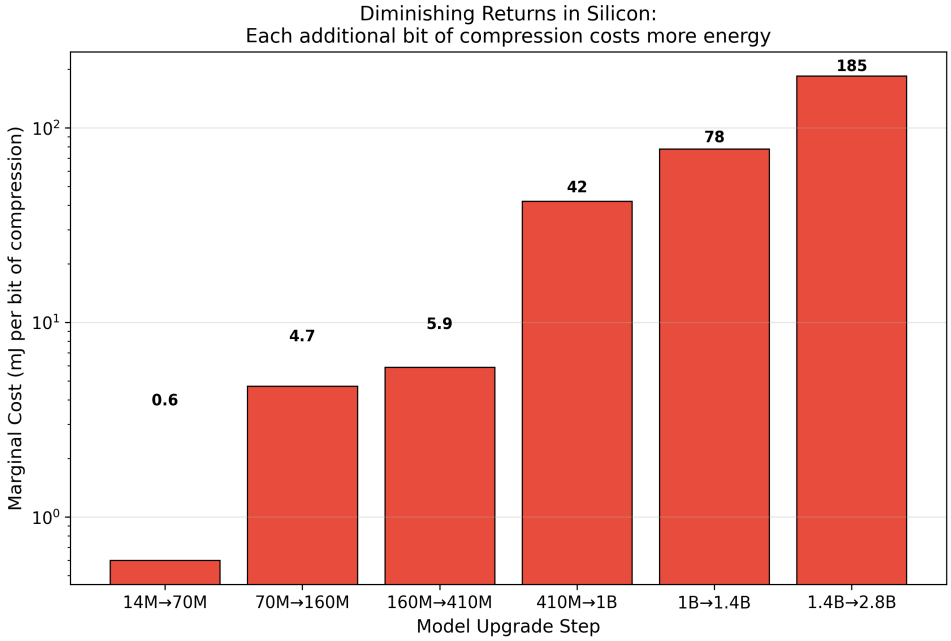

Figure 3. Diminishing Returns in Silicon. Marginal compression cost increases 300× across the Pythia family. Concave returns are universal; convex cost is substrate-dependent.

4.4 Slack Ordering

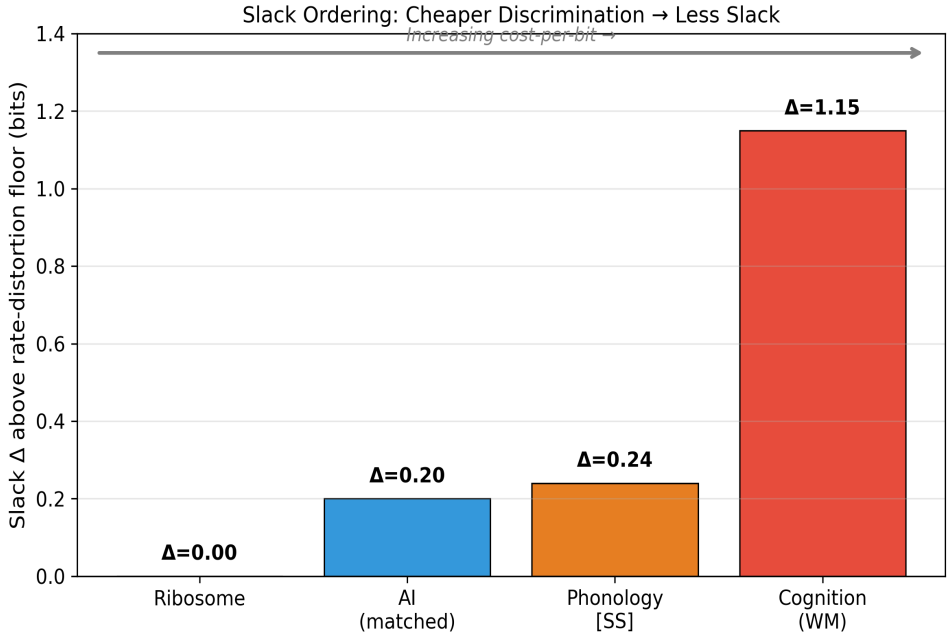

Figure 4. Slack Ordering (Regime A). Cheaper discrimination → less slack. This ordering applies within Regime A; Regime B slack is driven by model capacity.

5. Numerical Validation (Regime A)

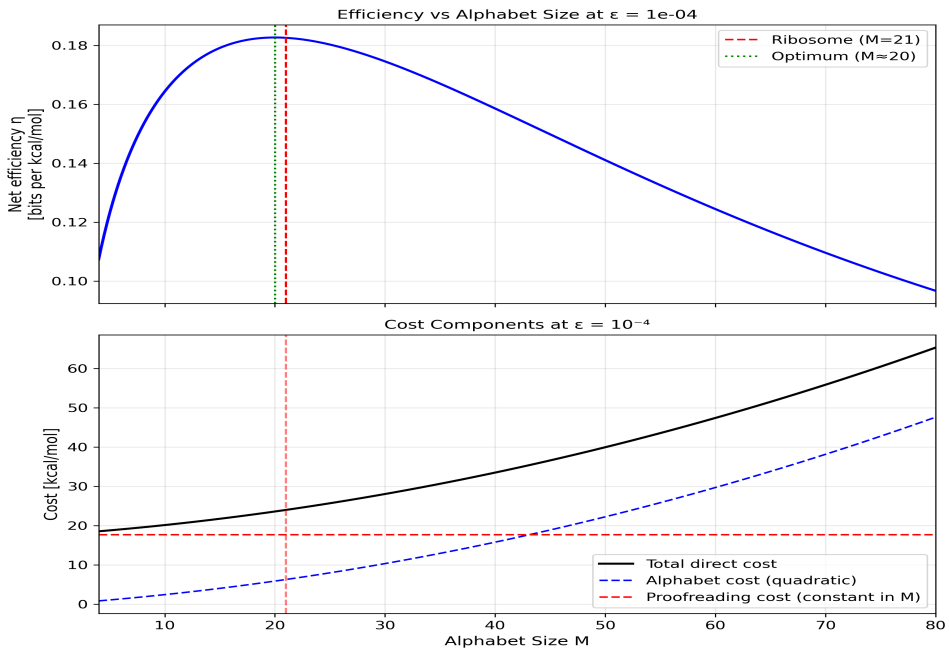

Figure 5. Efficiency vs. M at $\varepsilon = 10^{-4}$. Peak at M ≈ 20. This interior maximum exists because $\alpha_{alphabet} > 1$. In Regime B ($\alpha_{capacity} < 1$), no peak exists—efficiency decreases monotonically with scale.

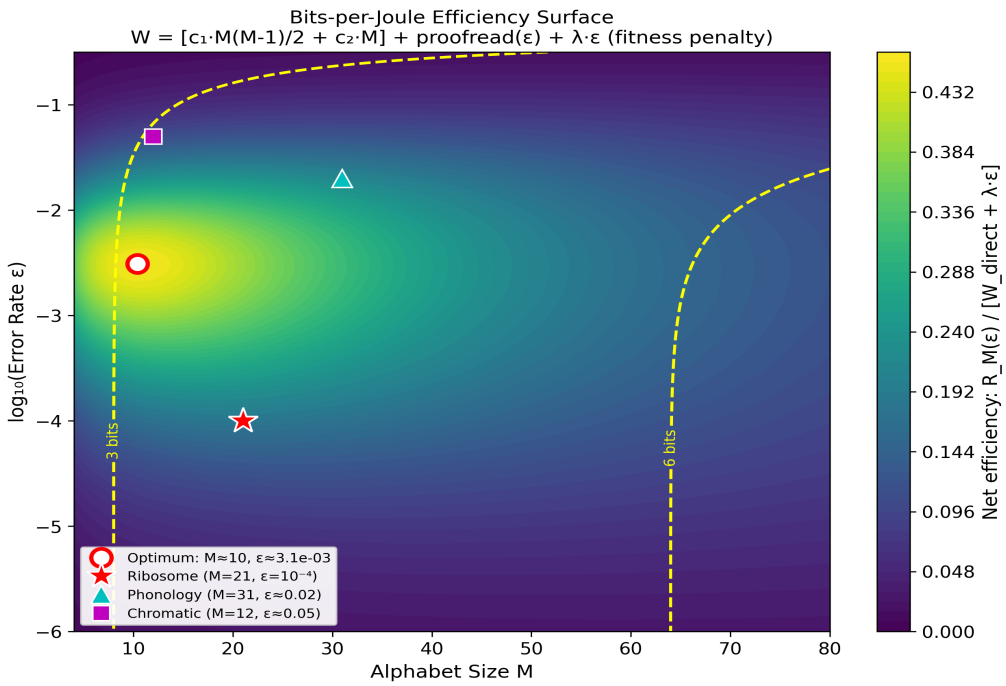

Figure 6. 2D Efficiency Surface (Regime A). Ribosome's M is explained by cost optimization; its ε reflects additional protein-misfolding fitness cost.

6. Dissipative Structure Interpretation

A Regime A serial decoder at $\tau \approx 4.2$ bits/event sustains informational order by continuously exporting entropy—a hallmark of dissipative structures (Prigogine, 1977). We use the term in its thermodynamic sense: a system maintained far from equilibrium by energy throughput. The

decomposition $I_{eff} = R_M(\varepsilon) + \Delta + \xi$ maps to: informational floor + thermodynamic slack + optimization residual. Regime B systems (AI) outsource entropy export to external infrastructure and are not self-contained dissipative structures in this sense.

7. Predictions and Empirical Tests

7.1 Temperature Dependence of Amino Acid Usage

Prediction: Discrimination cost increases with temperature ($W_{Landauer} \propto kT$). Thermophilic organisms should show reduced amino acid entropy.

Preliminary evidence (Kazusa-verified): Codon usage tables for 29 of 34 queried organisms were retrieved from the live Kazusa Codon Usage Database. The raw correlation between H_{AA} and growth temperature is marginal ($r = -0.320$, $p = 0.091$). GC-content is a significant confound ($r = -0.539$, $p = 0.003$). The partial correlation controlling for GC% is $r_{partial} = -0.451$ ($p = 0.014$)—significant but moderate.

Prior work has documented temperature-driven amino acid compositional shifts attributed primarily to GC-content bias and protein thermostability (Singer & Hickey, 2003; Zeldovich et al., 2007; Basak et al., 2006). Our H_{AA} entropy channel captures a complementary signal: not which amino acids are preferred, but how concentrated usage becomes. The partial correlation suggests a residual temperature effect beyond what GC-content and stability constraints explain, but this must be confirmed by PGLS with phylogenetic controls before it can be considered established.

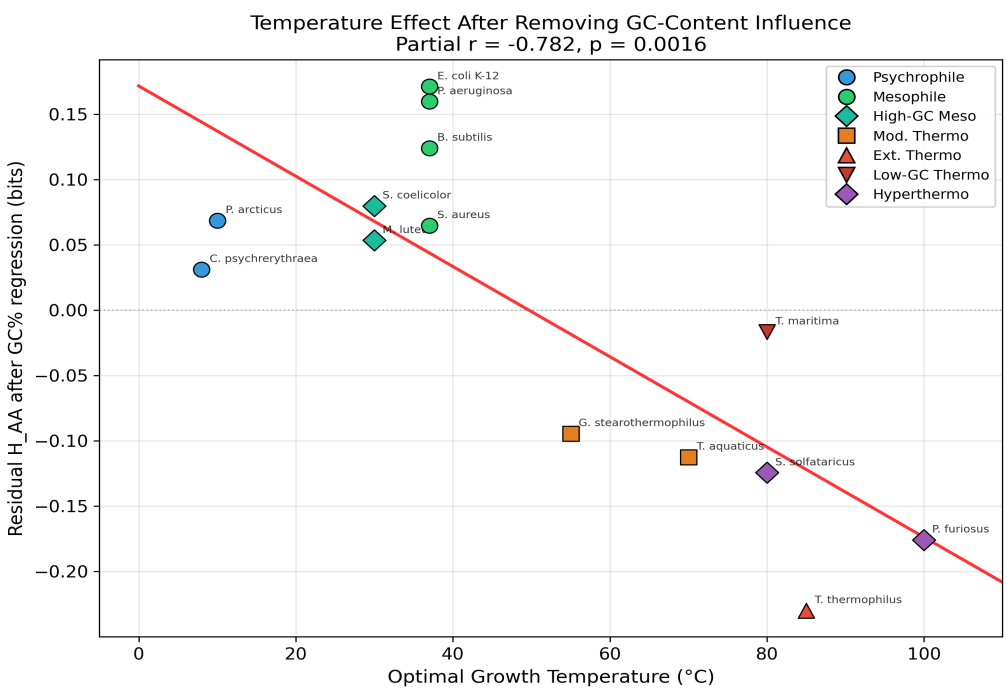

Figure 7. GC-Corrected Residuals vs. Temperature. Partial $r = -0.451$, $p = 0.014$ ($n = 29$, Kazusa-verified). Preliminary evidence pending phylogenetic controls (PGLS).

7.2 Silicon Energy: A Falsified Prediction and Its Resolution

Original prediction: Higher energy → higher slack (positive correlation). **Result: Falsified.** Spearman $\rho = -0.854$ ($p < 0.0001$, $n = 27$). Diagnosis: the scaling law dominates (larger models

are simultaneously more expensive and better at compression). Resolution: the prediction conflated $\alpha_{capacity}$ with $\alpha_{alphabet}$. Silicon's sub-linear capacity scaling means no cost-driven basin exists. AI's throughput near ~4.2 bits/token reflects training data statistics, not thermodynamic cost (§8.1).

7.3 Expanded Genetic Codes (Untested)

Prediction: GTP cost per codon increases super-linearly with M > 21. This is a direct test of $\alpha_{alphabet} > 1$ in biology. **Falsification:** Linear or flat cost increase.

8. Discussion

8.1 Two Regimes, One Mathematics

The throughput basin is a conditional attractor: inevitable for Regime A (alphabet-bound, $\alpha_{alphabet} > 1$), absent for Regime B (capacity-bound, $\alpha_{capacity} < 1$). The mathematics is unified—the same efficiency ratio $\eta = R/W$ governs both—but the physical instantiation differs. Biology *could not* decouple vocabulary from discrimination cost because molecular recognition is inherently pairwise. Silicon *could* because matrix multiplication does not require pairwise comparison. The Regime A/B distinction is the paper's central insight: it explains both where the basin exists (biology) and where it does not (silicon).

8.2 The AI-Dependence Problem: Resolved

Cross-domain clustering loses significance without AI data ($p = 0.56$; Whitmer, 2026c). This is now expected: AI does not belong in the Regime A basin. Its convergence near ~4.2 bits has a different cause—inherited training data structure from biological brains that ARE constrained by the basin. Removing AI from the Regime A analysis is correct.

8.3 Comparison with Prior Thermostability Hypotheses

Singer & Hickey (2003) and Zeldovich et al. (2007) documented amino acid compositional shifts in thermophiles, attributing them primarily to GC-content mutational bias and protein thermostability requirements. Basak et al. (2006) showed codon usage bias correlates with growth temperature. Our H_{AA} entropy measure captures a complementary signal: the overall *concentration* of amino acid usage, not just which amino acids are preferred. The partial correlation ($r = -0.451$ after GC correction) suggests the cost framework explains variance beyond what GC-content and stability constraints account for, but the effect is moderate and requires phylogenetic confirmation.

8.4 Limitations

Biological $\alpha_{alphabet} \approx 2$ is the theoretical prediction from pairwise combinatorics; direct measurement requires expanded-code translation experiments (Prediction 7.3). Silicon $\alpha_{capacity} = 0.937$ measures a different physical quantity than $\alpha_{alphabet}$; the direct quantitative comparison between regimes is not valid—only the qualitative distinction ($\alpha > 1$ vs. $\alpha < 1$) is claimed. The nvidia-smi benchmark measures empirical GPU energy consumption, not the Landauer minimum; the comparison to biology's $\phi \approx 1.02$ involves different metrics (useful dissipation fraction vs. raw power draw). Thermophilic analysis lacks phylogenetic controls (PGLS). Earlier AI-recalled codon usage values inflated the effect (recalled $r = -0.782$ vs. verified $r = -0.451$); all reported numbers use Kazusa-verified data.

9. Conclusion

The throughput basin is the intersection of a universal rate-distortion floor with a substrate-dependent cost surface. For Regime A systems—where discrimination cost scales super-linearly with alphabet size—the basin is a geometric necessity. The ribosome's $M \approx 21$ sits at the computed cost optimum. Amino acid entropy decreases with temperature as predicted (Kazusa-verified: partial $r = -0.451$, $p = 0.014$, pending PGLS).

For Regime B systems—where capacity cost scales sub-linearly and vocabulary is architecturally decoupled from discrimination—no cost-driven basin exists. Silicon operates $\sim 10^9\times$ above its Landauer floor; biology operates within 2%. The framework explains both: where the basin exists (biology) and how it can be escaped (silicon), through the architectural decoupling of vocabulary from pairwise recognition.

If confirmed by expanded-code calorimetry (Prediction 7.3) and phylogenetically controlled comparative genomics, the Regime A basin would constitute a substrate-specific thermodynamic bound on biological information processing: not a description of any particular organism, but a constraint on what any pairwise-discriminating system can achieve.

Appendix A: Derivations

A.1 ϕ_useful ≈ 1.02

4 GTP × 6 kcal/mol = 24 kcal/mol. Landauer: $kT \cdot \ln(2) \cdot I_{eff}$ = 0.00187 kcal/mol. η = 0.223 (Piñeros & Tlusty, 2020). $\phi = \eta \cdot n_c \cdot \Delta G$ / (RT ln 2) = 0.223×3×2.8 / (0.001987×310×0.693) = 4.387/4.270 = **1.027**. Predicted: 4.387 bits. Observed: 4.390. Δ = 0.003 bits.

A.2 Uniqueness of Maximum (Regime A)

When $\alpha_{alphabet}$ > 1: dR/dM = 1/(M ln 2) (decreasing); dW/dM = c■(M–1/2) + c■ (increasing). Quotient rule: $d\eta/dM$ changes sign exactly once → unique interior maximum. When $\alpha_{capacity}$ < 1: dW/dM is sub-linear (decreasing marginal cost); the ratio may decrease monotonically—no interior peak, no basin.

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Competing Interests: None. **Data:** GitHub + Zenodo. **License:** CC BY 4.0. **ORCID:** [pending].